

Deep Learning with Applications in Natural Language Processing

Lecture 1. From signals to tokens: basic text processing and beyond

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Course structure

Foundations

- Text processing and tokenization
- Statistical language models, subwords
- Syntax: constituency and dependency
- Sparse and dense representations

Architectures

- Feed-forward networks and CNNs
- Word embeddings, then RNNs
- Attention and Transformers

Modern NLP

- Pretraining and fine-tuning
- LLMs, prompting, RAG, agents
- Evaluation and responsible use
- Multimodal models

Evaluation

Component	What is assessed	Weight
Course	Written test	40%
Laboratory	Weekly lab work, first half of the semester	30%
Project	Progress in the second half and final presentation	30%

Use of generative AI follows the course policy and must remain auditable: students are responsible for checking claims, code and citations and for understanding AI's suggestions.

Today: seven questions

1. What does an NLP system actually receive: bytes, characters, tokens?
2. Where should token boundaries be placed, and what are tokens used for?
3. How can speech, images or sign language become tokens?
4. What information is inside a word: morphemes, stems, lemmas?
5. How do we assign a lexical class to a token in context?
6. What does “similar” mean under different representations?
7. What makes a corpus useful, or misleading, for a task?

The goal is not to memorize terminology, but to understand which decisions change what the model receives.

How do people communicate?

Speech

A continuous acoustic signal.

Gesture and sign

Gestures, signals, full sign languages.

Writing

Text in many scripts and writing systems.

Images and action

Pictures, diagrams, demonstrations.

Each channel is a different signal. To process it, a model first cuts it into **units**: tokenization in the broad sense.

Communication as a pipeline

Speaker S

- **Intention:** S wants H to believe P
- **Generation:** S chooses words W
- **Synthesis:** S utters W

Hearer H

- **Perception:** H perceives W'
- **Analysis:** possible meanings $P_1 \dots P_n$
- **Disambiguation:** S meant P_i
- **Incorporation:** believe P_i or not

Adapted from Russell & Norvig, Artificial Intelligence: A Modern Approach.

What is NLP?

A subfield of AI, close to computational linguistics: human-machine communication in natural language.

Understanding (NLU)

Language in, meaning out: classification, extraction, question answering.

Generation (NLG)

Meaning in, language out: translation, summarization, dialogue.

Large language models do both, but the distinction still matters for evaluation.

NLP applications

Classic

Machine translation,
retrieval, text classification,
information extraction, spell
checking

Today

Assistants, summarization,
retrieval-augmented
generation, code generation,
AI-text detection

Multimodal

Speech recognition and
synthesis, image captioning,
sign language translation

A running example

The European Union's Artificial Intelligence Act was published in the Official Journal on 12 July 2024 and entered into force on 1 August 2024. The regulation, proposed by the European Commission in April 2021, is described as the first comprehensive legal framework for AI. Its rules for general-purpose AI models have applied since 2 August 2025.

What does a machine need in order to understand this text?

Levels of analysis

Level	Question	In the example
Tokens	Where are the units?	Union's · 12 July 2024

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Syntax	How do words combine?	The ... Act was published
Semantics	Who did what, when?	Time: 1 August 2024
Discourse	What refers to what?	The regulation → the Act (anaphora)

What the machine actually sees

```
52 65 67 75 6C 61 6D 65 6E 74 75 6C 20 70 72 69 76 69 6E 64 20 69 6E 74 65 6C 69 67 65 6E C8 9B 61  
20 61 72 74 69 66 69 63 69 61 6C C4 83
```

“Regulamentul privind inteligența artificială”: 44 characters, **46 bytes** in UTF-8. The letters **ț** (C8 9B) and **ă** (C4 83) take two bytes each.

Encoding, normalization and diacritics are decided here, and they affect everything downstream.



PART 1

Tokenizing text

Breaking a character stream into the units a model works with.

Tokenization

Splitting a character stream into **tokens**: words, punctuation, numbers. A **type** is a distinct token.

Problem	Examples
Clitics	Union's · c'est · într-un · dă-mi-l
Numbers, dates	12 July 2024 · 2024/1689 · 3,14
Multi-word units	European Union · Official Journal
Web text	URLs · e-mails · #tags · emoji

How many tokens?

今天天气晴朗

it's sunny today

c'est ensoleillé aujourd'hui

Heute ist ein Sonnentag

- **Chinese:** no spaces; segmentation is a task
- **French:** two different apostrophes above
- **German:** compounds pack words together

Romanian-specific issues

Two ș letters

ș U+0219 (comma, correct) vs ș U+015F (cedilla). Same for ț / ț. Normalize first.

NFC vs NFD

“ă” as one code point, or “a” plus a combining breve. Same look, different bytes.

Clitics

dă-mi-l, s-a dus: one written word (space separated), several syntactic words.

Sentence segmentation

A period does not always end a sentence.

Reg. (EU) 2024/1689 was published on 12 July 2024. It applies, e.g., to GPAI models from 2 Aug. 2025.

- Abbreviations, initials, decimals, URLs
- Rule-based (abbreviation lists) or learned segmenters
- Speech has no punctuation: pauses (breaks) and prosody are used instead

The granularity spectrum

Unit	Idea
Words	whitespace and rules; large vocabulary, a lot of unknown words
Subwords	frequent words appear as a whole, rare words appear in pieces
Characters	tiny vocabulary, long sequences
Bytes	256 symbols, any script (ByT5)
Learned chunks	the model chooses the boundaries

Subwords are the standard in LLMs: no unknown words, reasonable sequence length.

How subword vocabularies are learned (BPE, WordPiece, Unigram): next lecture.

Example: one word, four granularities

Level	inteligența	Count
Word	[inteligența]	1
Subwords (e.g.)	[intelig] [ența]	2
Characters	i n t e l i g e n ț a	11
UTF-8 bytes	... 6E C8 9B 61	12

The subword split is illustrative: each tokenizer learns its own.

Tokenizer-free models

Entropy patches

BLT (ACL 2025): bytes grouped into patches; a new patch starts where the next byte is hard to predict.

Dynamic chunking

H-Net (ICLR 2026): segmentation learned end-to-end with the model, directly on bytes.

Results

BLT matches tokenizer-based LLMs at scale; a two-stage H-Net matches a token-based Transformer twice its size.

Pagnoni et al. 2025 · Hwang, Wang & Gu 2026. Comparisons hold in the authors' compute-matched settings.

Tokenization is not neutral

- The same text, translated, can need very different numbers of tokens: up to 15× between languages
- More tokens: higher cost, shorter usable context, slower output
- Diacritic variants (ș / ș) fragment words further
- **Exercise:** tokenize the working example in English and Romanian with two LLM tokenizers; compare tokens per word

Petrov et al. (2023), NeurIPS · Ahia et al. (2023), EMNLP · Rust et al. (2021): tokenizer quality affects downstream performance.

What are tokens used for?

Statistics

Word and sentence length,
type/token ratio,
frequency lists, style
identification

Language ID

Character n-gram profiles
reveal the language.

Input to other tasks

NER, POS tagging and
parsing label tokens,
using context.

Tokenization only provides the units.

Recognizing entities or categories is a separate, contextual task.

The running example

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Example: statistics of the working example

Measure	Value
Tokens (with punctuation)	61
Word tokens	56
Word types (lowercased)	45
Type/token ratio	$45 / 56 = 0.80$
Most frequent word	“the” (5 times)
Average word length	5.1 characters

Computed with a simple regular-expression tokenizer: `\w+` and single punctuation marks.

Example: language identification from characters

English profile

the · ing · and · ion

Romanian profile

ul_ · ții · ea_ · ște

Unknown text: *regulamentul european* contains **ul_** and **ean**: closer to the Romanian profile.

Illustrative trigrams. Real systems compare full n-gram frequency profiles (e.g. with cosine similarity).

Named Entity Recognition: a contextual task

Find names of real-world objects in text: persons, organizations, places, dates, laws, amounts.

NER labels a **sequence of tokens**, and the label of each token depends on its context:

Washington visited Iași. → PERSON

She moved to Washington. → LOCATION

Used for: information extraction, recommendation, anonymization (GDPR).

Surface cues are not enough

Token(s)	Cue	Entity?
European Commission	capitalized mid-sentence	yes: organization
12 July 2024	digits + month name	yes: date
The (sentence start)	capitalized	no
AI	all capitals	depends on context

Surface cues help, but only a model with context decides.

Example: NER as sequence labelling

The O
European B-ORG
Commission I-ORG
proposed O
the O

AI B-LAW
Act I-LAW
in O
April B-DATE
2021 I-DATE

BIO tags: B = beginning of an entity, I = inside, O = outside. Multi-word entities stay together: *European Commission* is one ORG.



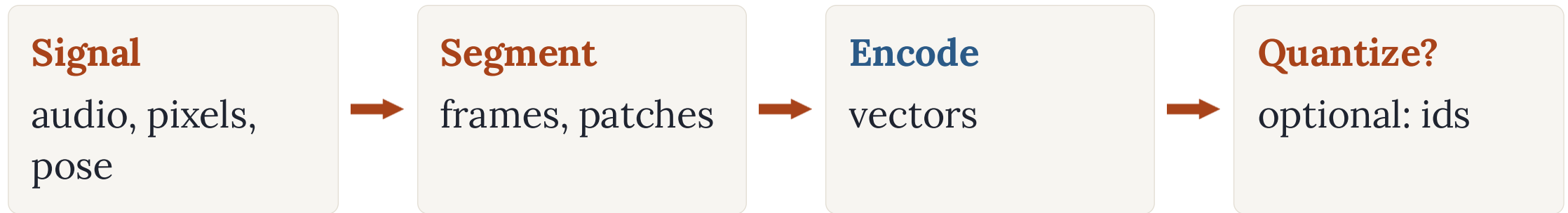
PART 2

Tokenizing other modalities

Speech, music, images, video, sign language and robot actions: continuous signals turned into sequences a model can predict.

What is a token, in general?

A unit in the sequence a model processes. Depending on the architecture, it is a **discrete id** or a **continuous vector**.



Text tokenization is reversible only without normalization (lowercasing or NFC lose information). Signal tokenization with quantization is always lossy.

Example: how many tokens?

Input	Raw size	Units
1 s of speech (24 kHz)	24,000 samples	12.5 frames $\times$ 8 codebook ids = 100 ids
224 $\times$ 224 RGB image	150,528 values	196 ViT patches; 32 with TiTok
Our example text	348 characters	61 word-level tokens

Discrete speech tokens

	Acoustic tokens	Semantic tokens
From	neural audio codecs	self-supervised models
Goal	rebuild the waveform	capture phonetic content
Examples	EnCodec, DAC, Mimi	HuBERT units, S3
Keeps	voice and prosody	content, at low bitrate
Loses	compactness	speaker detail

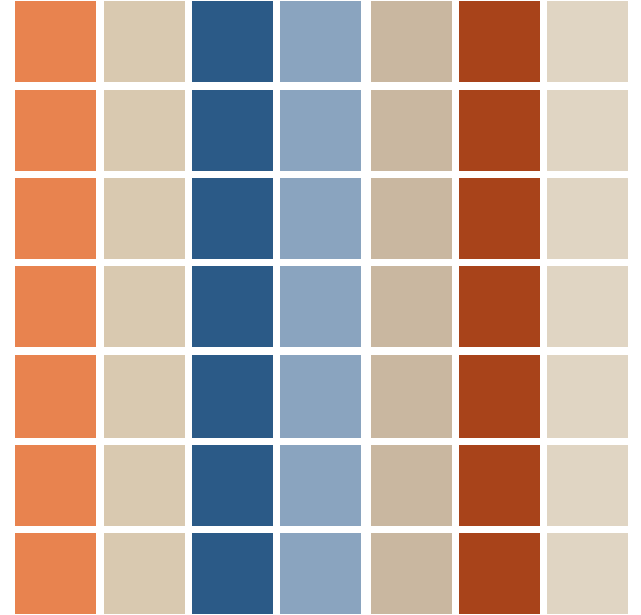
Guo et al. (2026), Recent Advances in Discrete Speech Tokens, IEEE TPAMI.

Images: patches as tokens

Vision Transformer: the image is cut into 16×16 -pixel patches, each projected to a vector.

A 224×224 image gives $14 \times 14 = \mathbf{196}$ tokens.

These tokens are continuous. Many vision-language models feed them directly into the LLM.



Dosovitskiy et al. 2021, An Image Is Worth 16×16 Words

Sign languages are natural languages

- Own grammar, lexicon and dialects
- Space used for reference and agreement
- Parallel channels: hands, face, mouth, body

Why tokenizing is hard

No common written form, simultaneous channels, little annotated data.

Sign language: symbolic notations

Glosses

Words for signs; lossy.

HamNoSys

Phonetic transcription.

SignWriting

Visual script; FSW is linear.

ASL “Hello”, FSW:

M518x529S14c20481x471S27106503x489

Tokens (vocabulary of 1,182):

M p518 p529 S14c c2 r0 p481 p471

S271 c0 r6 p503 p489

Summary across modalities

Modality	Unit	Method
Text	subword	BPE, Unigram
Speech	codec / SSL unit	RVQ, k-means
Music	event / codec	REMI, EnCodec
Image	patch / code	ViT, VQGAN, TiTok
Video	space-time code	MAGVIT-v2, Cosmos
Sign	symbol / motion	FSW, VQ on pose
Actions	frequency code	FAST



PART 3

Morphology

Stems, lemmas and the information inside words.

Computational morphology: analysis

Words are built from **morphemes**: a root plus affixes, e.g. **planning**, **redo**, **unbelievable**.

stopping $\Rightarrow$ stop+Verb+Cont

happiest $\Rightarrow$ happy+Adj+Superl

went $\Rightarrow$ go+Verb+Past

books $\Rightarrow$ book+Noun+Pl

book+Verb+Pres+3Sg

Computational morphology: generation

Given a lemma and features, produce the word form, e.g. in machine translation.

stop+Past $\Rightarrow$ stopped

cânta+PastPerf+1Pl $\Rightarrow$ cântaserăm

cânta+PastPerf+2Pl $\Rightarrow$ cântaserăți / cântaseți

PastPerf = Romanian mai-mult-ca-perfect.

Stemming vs lemmatization

Stemming

Rewrite rules (sses → ss). The stem need not be a word.

Word	Stem
connections	connect
university	univers
universe	univers

Lemmatization

Dictionary form; needs analysis and a lexicon.

Form	Lemma
studies	study
are	be
is	be

“studies” is ambiguous: noun plural or verb 3rd sg.

Example: lemmatizing Romanian

Form	Analysis	Lemma
copiilor	noun, pl., gen./dat., definite	copil
mergeam	verb, imperfect	merge
frumoasele	adjective, fem. pl., definite	frumos
fete	noun, plural	fată

Vowel alternations (fată / fete, frumos / frumoasele) make simple suffix-stripping fail.

Quick activity: what should be normalized?

Input	Possible operation	Risk
Universităţii	lemmatize	loses the case and number a task may need
ŞTIINŢĂ	lowercase	loses capitalization cues used by NER
copiilor	stem	may produce an opaque stem
ştiinţă	Unicode normalize (ș → ş)	usually desirable before comparison

Two minutes in pairs: for a search engine, a spell checker and an NER system, which operations would you apply?

Decide per task, not by habit.



PART 4

Part-of-speech tagging

Assigning each word its lexical class. Sentence structure follows in Lecture 3.

Part-of-speech tagging

Assigning a lexical class to every token, in context.

the	koala	put	the	keys	on	the	table
DET	NOUN	VERB	DET	NOUN	ADP	DET	NOUN

I like *research* (noun) vs I *research* NLP (verb).

Example: POS ambiguity in Romanian

O	văd	cum	poartă	o	poartă	grea
PRON	VERB	CONJ	VERB	DET	NOUN	ADJ

The same form **poartă** is a verb (“wears”) and a noun (“gate”). The context decides.

o can also be a numeral (“one”) or a pronoun (“o văd”, “I see her”).

Tagsets

Penn Treebank

36 tags plus punctuation, English only: NN, NNS, VBD, JJR ...

Universal Dependencies POS

17 tags for 100+ languages:

ADJ ADP ADV AUX CCONJ DET INTJ NOUN NUM PART
PRON PROPN PUNCT SCONJ SYM VERB X

“first” is an ordinal: JJ in Penn, ADJ in UD.

Tools: spaCy, Stanza, UDPipe.

Why POS tags still matter

Speech synthesis

OBject / obJECT; l
ead (verb) vs lead (metal).

Disambiguation

Time flies like an arrow:
flies, is it a noun or a verb?

Probing

Diagnostic targets for
what neural models learn.

Next: from words to sentences

POS tags are the input to **Lecture 3**, which covers sentence structure:

Constituency

```
[S [NP Ea]
  [VP poartă
    [NP o poartă grea]]]
```

Dependency

```
poartă → Ea (nsubj)
poartă → poartă (obj)
```

Syntactic ambiguity, parsing algorithms and semantic roles are all in Lecture 3.



PART 5

Similarity

Strings, vectors, sets and knowledge bases.

Similarity in NLP

Task	Examples
Text similarity	plagiarism, FAQ matching, retrieval
Example-based learning	nearest-neighbour tagging, NER
Classification	sentiment, spam, routing
Evaluating generation	MT or summaries vs references
Alias detection	President Hennessy = John Hennessy

Distance, similarity, relatedness

Distance (a metric)

$$d(x,x) = 0$$

$$d(x,y) > 0 \text{ if } x \neq y$$

$$d(x,y) = d(y,x)$$

$$d(x,z) \leq d(x,y) + d(y,z)$$

Similarity

sim: $A \times B \rightarrow [0,1]$. Need not be symmetric or obey the triangle inequality.

$$\text{sim} = 1 / (1 + d) \quad d = 1 / \text{sim} - 1$$

$$d = 0 \rightarrow \text{sim} = 1; \quad d = 1 \rightarrow \text{sim} = 0.5; \quad d = 3 \rightarrow \text{sim} = 0.25$$

Approaches to similarity

Approach	Idea
I. String	edit distance (Levenshtein)
II. Vector	Euclidean, dot product, cosine
III. Set	Jaccard
IV. Knowledge	paths in WordNet
V. Corpus	distributional vectors

Adapted from L. Padró, Similarity models in NLP (UPC).

I. Edit distance

Fewest edits that turn one sequence into another.

- **LCS:** insert, delete
- **Levenshtein:** + substitute
- **Damerau:** + transpose

kitten → sitten
sitten → sittin
sittin → sitting
distance = 3

Used in : spelling, MT evaluation, word error rate.

Levenshtein: dynamic programming

Levenshtein: SUNDAY → SATURDAY

	λ	S	A	T	U	R	D	A	Y
λ	0	1	2	3	4	5	6	7	8
S	1	0	1	2	3	4	5	6	7
U	2	1	1	2	2	3	4	5	6
N	3	2	2	2	3	3	4	5	6
D	4	3	3	3	3	4	3	4	5
A	5	4	3	4	4	4	4	3	4
Y	6	5	4	4	5	5	5	4	3

Distance = **3**: insert A, insert T, substitute N → R.

Edit distance over words

Spokesman → The substitute
confirms → spokesman substitute
 + said insert
 + the insert
senior = senior
government – delete
advisor was shot = (3 matches)
 + dead insert

Distance = **6** when case counts (Spokesman ≠ spokesman), **5** after lowercasing.

Sentence 1: Spokesman confirms senior government advisor was shot.

Sentence 2: The spokesman said the senior advisor was shot dead.

II. Vector similarity: three measures

Measure	Formula	Looks at
Euclidean distance	$\sqrt{\sum (x_i - y_i)^2}$	position
Dot product	$\sum x_i y_i$	angle and length
Cosine similarity	$x \cdot y / (x y)$	angle only

Example: $x = (1, 0)$, $y = (3, 0)$: cosine = 1 (same direction), dot product = 3, Euclidean distance = 2.

Dot product and cosine come back with embeddings and attention. More measures in the appendix.

Vector example

s1 *Spokesman confirms senior government advisor was shot*

s2 *The spokesman said the senior advisor was shot dead*

s3 *Spokesman said the shot government advisor was dead*

	Euclidean	Dot product	Cosine
s1-s2	2.83	5	0.57
s1-s3	2.24	5	0.67
s2-s3	1.73	8	0.85

Word-count vectors; “the” occurs twice in s2. s2 and s3 are closest by every measure.

III. Set similarity: Jaccard

$$\text{Jaccard}(X, Y) = |X \cap Y| / |X \cup Y| \quad \text{distance} = 1 - \text{Jaccard}$$

Treat each text as a **set** of words: only presence counts, not frequency.

Related: Dice = $2|X \cap Y| / (|X| + |Y|)$. Overlap and set cosine in the appendix.

Set example

	Jaccard	Dice
s1-s2	0.50	0.67
s1-s3	0.50	0.67
s2-s3	0.78	0.88

$s1 \cap s2 = \{\text{spokesman, senior, advisor, was, shot}\}$: 5 words; $s1 \cup s2$ has 10, so Jaccard = $5/10 = 0.50$.

IV. Symbolic knowledge: WordNet

- **Synsets:** synonym sets for one concept
- Relations: hypernymy (is-a), meronymy (part-of), antonymy
- car: {car, auto, automobile} and {car, railcar}
- Similarity = closeness in the graph

Today

Open English WordNet; RoWordNet for Romanian.

Example: dog vs cat in WordNet

Measure	dog.n.01 vs cat.n.01
Path similarity	0.20
Leacock–Chodorow	2.03
Wu–Palmer	0.86

```
from nltk.corpus import wordnet as wn
dog, cat = wn.synset('dog.n.01'), wn.synset('cat.n.01')
dog.wup_similarity(cat)
```

Values from NLTK with WordNet 3.0.

V. Distributional similarity

You shall know a word by the company it keeps. (Firth, 1957)

Symbolic (WordNet)

Meaning written down by experts: precise, but limited and costly.

Distributional

Meaning learned from contexts in a corpus: broad and automatic.

Sparse (term–document, PMI, TF-IDF) now; dense embeddings later in the course.

Example: a tiny term-document matrix

	d1	d2	d3
AI	1	2	0
models	1	1	1
generate	1	0	1
text	1	0	1
Act	0	1	0
language	0	0	1

d1: AI models generate text
d2: the AI Act regulates AI models
d3: models of language generate text

Cosine: $d1-d3 = 0.75$, $d1-d2 = 0.61$, $d2-d3 = 0.20$.

Function words (the, of) and “regulates” omitted.



PART 6

Corpora

The data every method in this course depends on.

What is a corpus?

Machine-readable, authentic texts, sampled to represent a language or variety.

After McEnery, Xiao & Tono (2006); Leech (1992).

Can

Test hypotheses; show what is typical; train models.

Cannot

Give negative evidence or explain what it shows.

Kinds of corpora

Dimension	Types
Modality	text, speech, video, sign language
Scope	specialized vs reference
Time	synchronic vs diachronic
Languages	monolingual; parallel; comparable
Scale	web-scale: Common Crawl, FineWeb
Romanian	CoRoLa, UD treebanks, RoWordNet

New concerns: licensing, benchmark contamination, documentation.



**Most deep learning needs a corpus.
A corpus is only useful if it fits the problem.**

References (1/2): foundations

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Thank you!

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PART A

Appendix

Details, catalogues and extra examples for reference and for the lab.

Speech before tokens: frames and features

Frames

Windows of about 25 ms,
every 10 ms.

Features

MFCC or log-mel
spectrogram (Whisper: 80
mel bins; 128 in large-v3).

Linguistic units

Phonemes, syllables, words:
need labels.

Spectrogram frames are continuous. To generate speech with an LLM we need discrete tokens.

From tokens to spoken language models

Hierarchy

AudioLM: semantic tokens first, then acoustic tokens, then audio.

Shorter sequences

Deduplicate repeated units; apply BPE on units, as for text.

Open issues

Variable frame rate; separating content, speaker, emotion.

Borsos et al. (2023), AudioLM · Mousavi et al. (2025), Discrete Audio Tokens, TMLR.

Example: semantic units for one word

frames: 71 71 71 12 12 45 45 45 45 88 88

dedup: 71 12 45 88

BPE: (71 12) 45 88

Each number is a cluster of HuBERT features (a k-means unit), roughly a phone-like sound. Repeats come from frames shorter than sounds.

Illustrative unit ids.

Music: symbolic or audio tokens

Symbolic (MIDI)

REMI events: bar, position, pitch, velocity, duration.

Audio

Codec tokens, used by text-to-music models.

Bar · Position · Note-Velocity · Note-On · Note-Duration · ...

Huang & Yang (2020), Pop Music Transformer, ACM Multimedia.

Images: discrete visual tokens

Tokenizer	Idea
VQ-VAE (2017)	learned codebook over a 2D grid
VQGAN (2021)	adversarial loss: sharper images
TiTok (2024)	1D tokens, as few as 32 per image
Semantic	tokens aligned with meaning

Trade-off: faithful pixels or semantic abstraction.

Video: space and time

3D tokens

MAGVIT-v2: lookup-free quantization, one codebook for images and video.

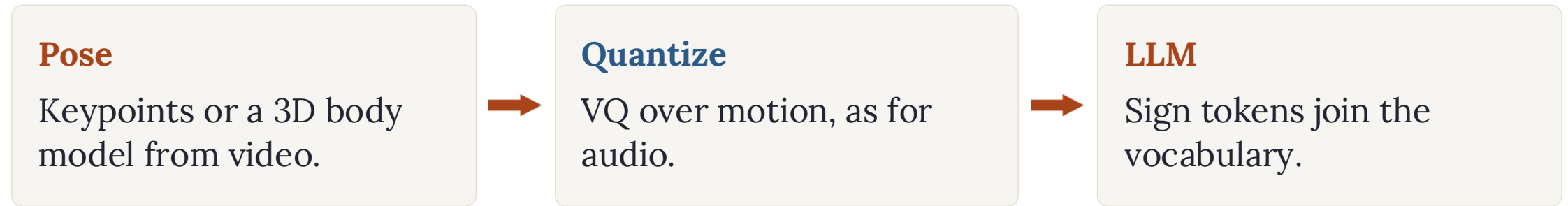
Causal

Cosmos Tokenizer (NVIDIA, 2025): keeps frame order; FSQ tokens.

Redundancy

Good tokenizers compress what does not change between frames.

Sign language: learned tokens from video



- Abzaliev & Mihalcea (EMNLP 2024): audio compression for ASL fingerspelling
- SignLLM (CVPR 2024): sign tokens aligned with text for a frozen LLM
- SOKE (ICCV 2025): a decoupled tokenizer with separate tokens for different body parts

Actions and time series

Robot actions: FAST

DCT to frequency space, round the coefficients, then **BPE**. Replaces per-step binning.

Time series: Chronos

Scale the series, quantize values into bins, train a language model on bin ids.

Same recipe: normalize, transform, quantize, compress.

Pertsch et al. 2025 · Ansari et al. 2024

Example: tokenizing a time series

Step	Values
Series	10, 12, 11, 30
Scale by the mean (15.75)	0.63, 0.76, 0.70, 1.90
Bins of width 0.5 → ids	1, 1, 1, 3

A language model then predicts the next bin id, as if it were a word.

Illustrative bin width; Chronos uses many more bins.

Other “languages” with tokens

Domain	Units
Molecules	SMILES or SELFIES strings
Proteins	amino acids; structure alphabets
Code	subwords plus indentation tokens
Recommendation	semantic IDs (code tuples)

Every SELFIES string decodes to a valid molecule (Krenn et al. 2020).

Design trade-offs

Trade-off	In short
Fidelity vs meaning	detail-rich tokens vs LLM-friendly tokens
Rate vs length	better quality, longer sequences
Codebook size	too small loses data, too large goes unused
Discrete vs continuous	VLMs often skip quantization
Unified vocabulary	early fusion, e.g. Chameleon (2024)

Appendix: more vector distances

Measure	Formula
Manhattan, L_1	$\sum x_i - y_i $
Euclidean, L_2	$\sqrt{\sum (x_i - y_i)^2}$
Minkowski, L_r	$(\sum x_i - y_i ^r)^{1/r}$
Chebyshev, L_∞	$\max_i x_i - y_i $ (limit of L_r)
Canberra	$\sum x_i - y_i / (x_i + y_i)$
Cosine (sim.)	$x \cdot y / (x y)$
Dot product (sim.)	$\sum x_i y_i$

Appendix: all set measures

	Dice	Jaccard	Overlap	Cosine
s1-s2	0.67	0.50	0.71	0.67
s1-s3	0.67	0.50	0.71	0.67
s2-s3	0.88	0.78	0.88	0.88

Overlap = $a / \min(a+b, a+c)$ Cosine = $a / \sqrt{(a+b)(a+c)}$

Appendix: distances in WordNet

Measure	Definition
Shortest path	$SLP(s_1, s_2)$
Leacock–Chodorow	$-\log(SLP / (2 \cdot \text{MaxDepth}))$
Wu–Palmer	$2 \cdot \text{depth}(\text{LCS}) / (\text{depth}(s_1) + \text{depth}(s_2))$
Gloss overlap	set or vector measure on definitions

SentiWordNet: each synset gets three scores (positive, negative, objective) that sum to 1.

Romanian: orthographic words ≠ syntactic words

Written form	Possible analysis	Syntactic words
dă-mi-l	dă + mi + l	verb + 2 clitic pronouns
s-a dus	s + a + dus	reflexive + auxiliary + verb
într-un	într- + un	preposition + article

There is no universally correct split outside a task and an annotation convention (e.g. UD Romanian).

Keep the original text whenever reversibility matters.

Appendix references: vision, sign language, other modalities

- van den Oord et al. (2017). Neural Discrete Representation Learning. NeurIPS · Esser, Rombach & Ommer (2021). Taming Transformers. CVPR
- Yu et al. (2024). An Image is Worth 32 Tokens for Reconstruction and Generation. NeurIPS · NVIDIA (2025). Cosmos Tokenizer
- Moryossef & Jiang (2023). SignBank+. arXiv preprint · Abzaliev & Mihalcea (2024). Unsupervised Discrete Representations of American Sign Language. EMNLP
- Gong et al. (2024). LLMs are Good Sign Language Translators. CVPR · Zuo et al. (2025). Signs as Tokens. ICCV

Appendix references: actions, time series, music, other

- Pertsch et al. (2025). FAST: Efficient Action Tokenization for Vision-Language-Action Models. RSS
- Ansari et al. (2024). Chronos: Learning the Language of Time Series. TMLR
- Huang & Yang (2020). Pop Music Transformer. ACM Multimedia · Borsos et al. (2023). AudioLM. IEEE/ACM TASLP
- Krenn et al. (2020). SELFIES. *Mach. Learn.: Sci. Technol.* · Rajput et al. (2023). TIGER. NeurIPS · Chameleon Team (2024). arXiv

Unicode normalization: same look, different code points

NFC (composed)

ă as one precomposed character, U+0103.

ă → C4 83

NFD (decomposed)

a followed by a combining breve, U+0306.

a + U+0306 → 61 CC 86

Visually equivalent is not byte-identical: compare and tokenize only after normalizing. NFC is the usual safe default.

NFKC / NFKD also remove compatibility distinctions (e.g. ligatures, superscripts) and should not be applied blindly. Unicode Standard Annex #15.